

List of Questions of Slide 6

1. Define Network Congestion?



Congestion (overloaded condition in a region of a network) may occur if the number of datagrams sent by source computers is beyond the capacity of the network or routers. In this situation, some routers may drop some of the datagrams.

2. Compare logical and physical congestion

Comparison Between Logical and Physical Congestion

Aspect	Logical Congestion	Physical Congestion
Definition	Congestion occurs due to excessive data packets in the network, leading to delays or packet loss.	Congestion occurs due to physical limitations such as insufficient bandwidth, hardware constraints(সীমাবদ্ধতা), or network failures.
Cause	Caused by high traffic load, inefficient routing, or protocol inefficiencies.	Caused by hardware failures, limited network capacity, or broken links.
Nature	Happens at the network layer or higher due to excessive logical data flow.	Happens at the physical layer due to hardware limitations or physical obstructions.
Solution	Can be managed using traffic control mechanisms like congestion control algorithms (e.g., TCP congestion control).	Requires hardware upgrades, increased bandwidth, or network infrastructure improvements.
Examples	Packet queuing, buffer overflow, TCP retransmissions due to congestion.	Overloaded routers, broken cables, slow transmission speeds due to physical limitations.

Key Takeaways

- **Logical congestion** results from excessive data flow and inefficient network handling.
- **Physical congestion** stems from hardware limitations and physical network constraints.
- **Logical congestion** can often be resolved by software-based solutions, while **physical congestion** requires hardware upgrades or infrastructure expansion.

3. Explain Operations of Load Shedding to Relieve Datagram Congestion

Load shedding is a congestion control mechanism where a router discards packets when it becomes overwhelmed with traffic. Instead of allowing excessive packets to degrade the entire network, the router selectively drops packets to maintain performance.

Key Operations of Load Shedding:

1. Packet Discarding Strategy:

- a. The router decides **which packets to drop** based on the type of application and network conditions.
- b. Instead of random dropping, it prioritizes discarding packets in a way that minimizes overall retransmission overhead.

2. Application-Specific Discard Policies:

a. For file transfer applications (e.g., FTP, HTTP downloads):

- i. Old packets are **more valuable** than new ones because losing an earlier packet can trigger retransmission of multiple subsequent packets.
- ii. **"Wine Policy"** → Old packets are **better** than new ones (keep older packets and drop newer ones).

b. For multimedia applications (e.g., VoIP, video streaming):

- i. New packets are **more valuable** than old ones because late-arriving packets become useless due to real-time playback constraints.
- ii. **"Milk Policy"** → New packets are **better** than old ones (drop older packets and keep newer ones).

3. Random Early Detection (RED) Mechanism:

- a. Some routers use **Random Early Detection (RED)** to proactively drop packets before congestion becomes severe, helping to prevent packet loss spikes.

4. Weighted Fair Queueing (WFQ):

- a. Ensures **priority traffic** (like VoIP) gets through while non-urgent traffic (like bulk downloads) can be delayed or dropped.

Summary:

- Load shedding helps relieve **network congestion** by **intelligently dropping packets** when the router buffer overflows.
- Different applications require **different strategies**:
- **File transfers** prioritize old packets (**Wine Policy**).
- **Real-time multimedia** prioritizes new packets (**Milk Policy**).
- Efficient load shedding **prevents network collapse** and **optimizes packet delivery** for critical applications. 🚀

4. Take you decision about text and multimedia traffic

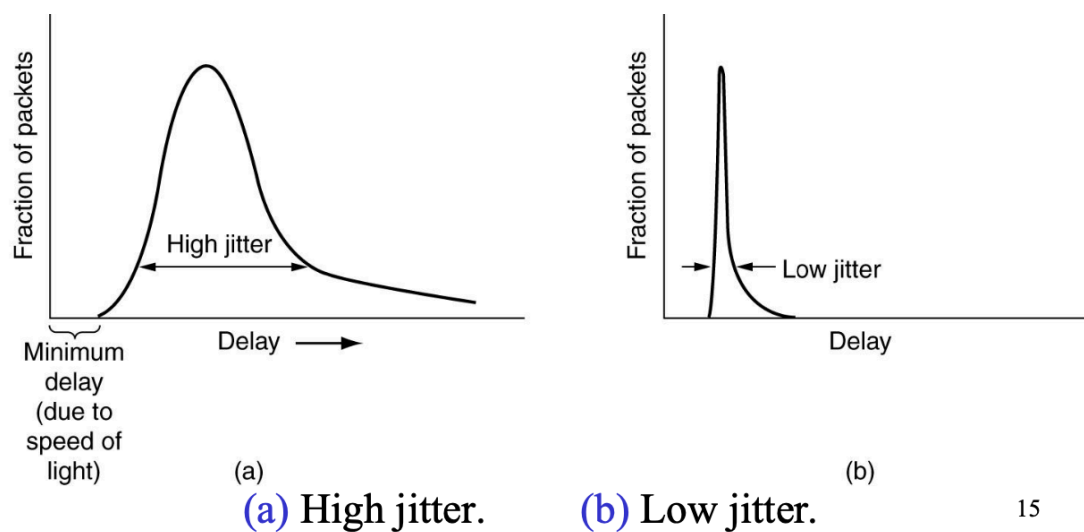
Define Jitter

Jitter Definition:

Jitter refers to the **variation in packet arrival time** when data is transmitted over a network. It occurs when packets take different paths or experience varying levels of congestion, leading to inconsistent delays. High jitter can cause **choppy audio, lag in video calls, and reduced network performance**.

For applications such as audio and video streaming, it does not matter much if the packet takes 20 ms or 30 ms to be delivered, as long as the transmit time is constant. The variation (standard deviation) in the packet arrival times is called jitter. High jitter, for example, having some packets taking 20 ms and others

taking 30ms to arrive will give an uneven quality of sound or movie. Jitter is illustrated graphically below.



15

The image illustrates **jitter** by comparing two cases: **high jitter** and **low jitter** in a network.

(a) High Jitter:

- The **x-axis represents delay**, and the **y-axis represents the fraction of packets** received with that delay.
- In this case, packets experience **large variations in delay**, meaning they arrive at inconsistent intervals.
- This is **problematic for real-time applications** like **VoIP, video calls, and online gaming** because some packets arrive too late to be useful.
- Causes of **high jitter**:
- Network congestion
- Variable queuing delays
- Routing changes

(b) Low Jitter:

- Packets arrive with **minimal variation in delay**.
- The delay distribution is **narrow**, meaning packets arrive at almost **regular intervals**.
- This is **ideal for real-time applications**, as it ensures **smooth audio/video transmission**.

- **Low jitter networks** typically have:
- Proper **QoS (Quality of Service) settings**
- **Sufficient bandwidth** and minimal congestion
- **Efficient routing and traffic management**

Conclusion:

- **Lower jitter is always preferable**, especially for real-time communication.
- **Jitter buffers** can help mitigate high jitter, but they add some delay.
- **Network optimization** is necessary to ensure stable packet delivery times.



5. Compare email, file transfer , video on demand and video conferencing in control of four QoS parameters

Requirements of QoS

A stream of packets from a source to destination is called flow. The needs of each flow can be characterized by four primary parameters: **reliability**, **delay**, **jitter** and **BW**. Together these determines the QoS the flow requires. Some common applications with required QoS is shown below.

Application	Reliability	Delay	Jitter	Bandwidth
E-mail	High	Low	Low	Low
File transfer	High	Low	Low	Medium
Web access	High	Medium	Low	Medium
Remote login	High	Medium	Medium	Low
Audio on demand	Low	Low	High	Medium
Video on demand	Low	Low	High	High
Telephony	Low	High	High	Low
Videoconferencing	Low	High	High	High

25

6. Example:

Example-1

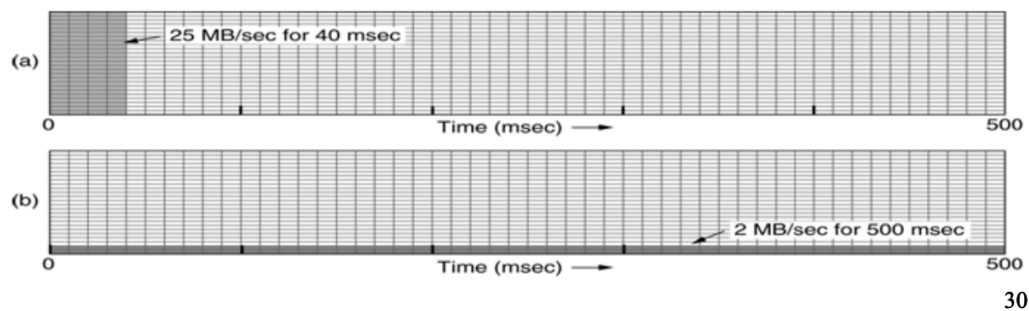
A node provides data burst at a rate $R = 25 \text{ MB/sec}$ for the duration of 40ms at the input of the leaky bucket. The output rate of the bucket is 2MB/sec. If the capacity of the buffer (or leaky bucket) is 10MB then determine (i) possible overflow (ii) time required to evacuate the bucket

Ans. (i) The amount of data burst enters the bucket is: $25 \times 40 / 1000 \text{ (MB/sec)} \times \text{sec} = 1 \text{ MB} < \text{Capacity of } 10 \text{ MB}$. Therefore no overflow of traffic.

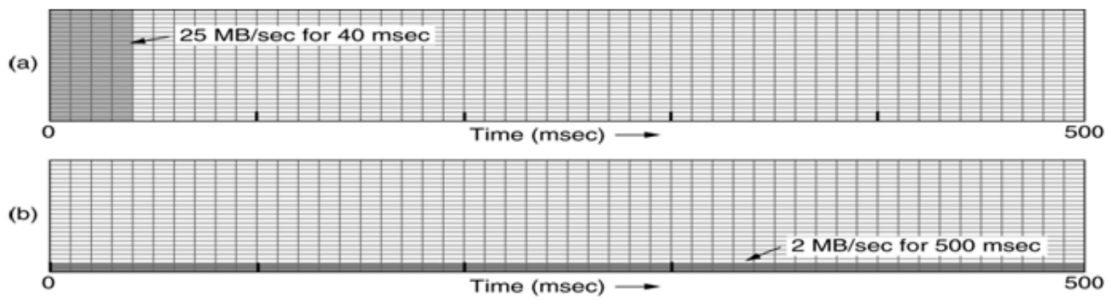
(ii) Let t be the time required to evacuate the bucket.

Now $C = R \times t$ or $1 = 2 \text{ MB/sec} \times t$ sec or $t = 0.5 \text{ sec} = 500 \text{ ms}$

The leaky bucket spread the burst over 500ms.



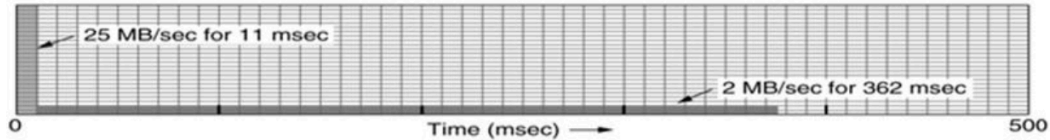
The input to the leaky bucket running at 25MB/sec for 40 ms. The output of the bucket draining out at a uniform rate of 2 MB/sec for 500ms.



(a) Input to a leaky bucket. (b) Output from a leaky bucket

7. Problem on token bucket

Calculating the length of the maximum rate burst is slightly tricky.
 The capacity of the token bucket $C = 250\text{KB}$ (Amount of reserved token)
 Token arrival rate, $\rho = 2\text{MB/sec} = 2000\text{KB/sec}$
 The maximum drain rate, $M = 25\text{MB/sec} = 25000\text{KB/sec}$
 Determine duration of both drain rate.



Let the duration of maximum rate is S .

Now, $M \cdot S = C + \rho \cdot S$

Or, $S = C / (M - \rho) = 250 / (25000 - 2000) \text{ sec} = 250 / 23000 \text{ sec}$
 $= 25 / 2300 \text{ sec} = 10.86 \text{ ms} = 11 \text{ ms}$

During 11ms only a fraction of the burst traffic will be released ($11 \text{ ms} \cdot 25 \text{ MB/sec} = 275 \text{ KB}$) at the rate M and the rest of the packet will be released at low rate $\rho = 2 \text{ MB/sec}$.

Here the volume of the burst is 1MB. Let the time required to drain the rest of the packet at a rate of $\rho = 2 \text{ MB/Sec}$ is t .

$1 \text{ MB} = (11/1000) \cdot 25 + 2 \cdot t$

Or, $t = 0.362 \text{ sec} = 362 \text{ ms}$

Consider a token bucket where token arrives at a rate of $\rho = 2\text{MB/sec}$. Assuming the token bucket is full when the 1MB burst arrives, the bucket can drain at the full rate $M = 25\text{MB/sec}$ against reserved token of $C = 250\text{KB}$.

During the draining of packet at 25MB/sec on the reserved token of $C = 250\text{KB}$, the bucket will get some additional token which also be used to drain at a rate of $\rho = 2\text{MB/sec}$.

$M \rightarrow$ High speed drain rate
 $\rho \rightarrow$ Low speed drain rate or token rate
 $C \rightarrow$ Reserve token inside the bucket
 $S \rightarrow$ Duration of high speed drain rate

$$M * S = C + \rho * S$$

$\uparrow$
 Total drained packet at
the rate of M during S

$\uparrow$
 Against the reserved
token of C

$\uparrow$
 Against the earned
token during S

Q. Consider the reserved token $C = 500\text{KB}$ and $M = 10\text{MB/sec}$. Determine duration of both drain rate.

Ans. The capacity of the token bucket $C = 500\text{KB}$

Token arrival rate, $\rho = 2\text{MB/sec} = 2000\text{KB/sec}$

The maximum drain rate, $M = 10\text{MB/sec} = 10000\text{KB/sec}$

Let the duration of maximum rate is S .

Now, $MS = C + \rho S$

$\Rightarrow S = C / (M - \rho) = 500 / (10000 - 2000) \text{ sec} = 500 / 8000 \text{ sec} = 5/80 \text{ sec}$
 $= 62.5\text{ms} = 62\text{ms}$

Here the volume of the burst is 1MB. Let the time required to maintain the token rate of 2MB/Sec to drain the entire burst is t .

$1\text{MB} = (62/1000) * 10 + 2 * t$

Or, $t = 0.19\text{sec} = 190\text{ms}$

